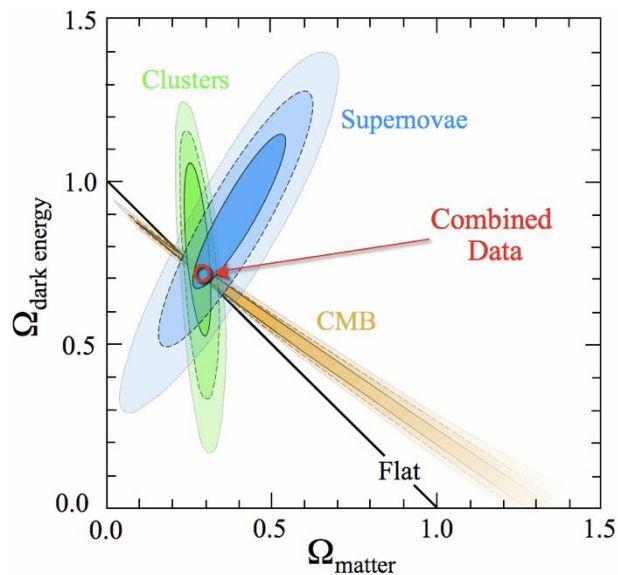
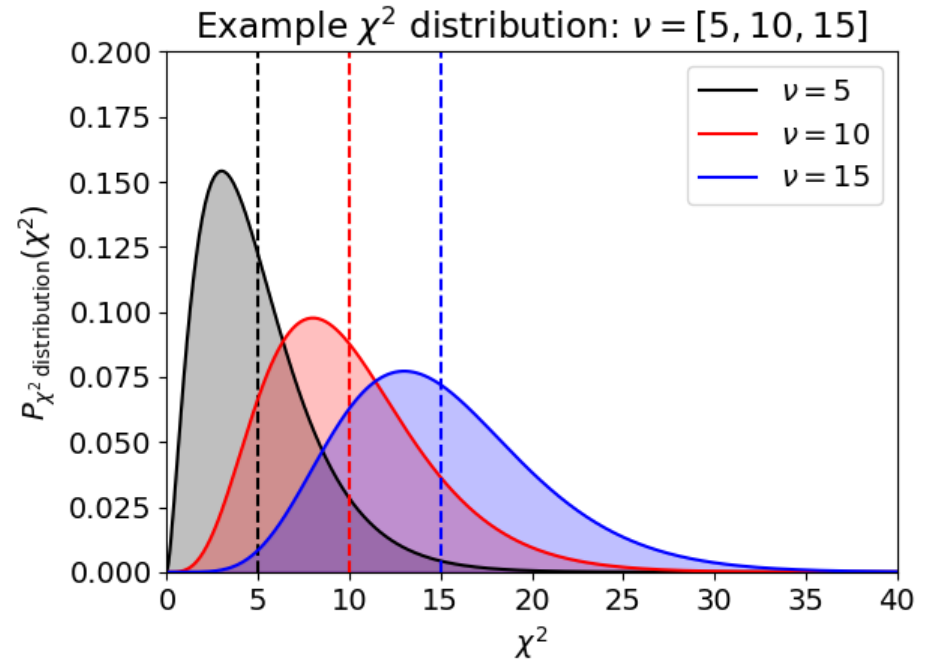
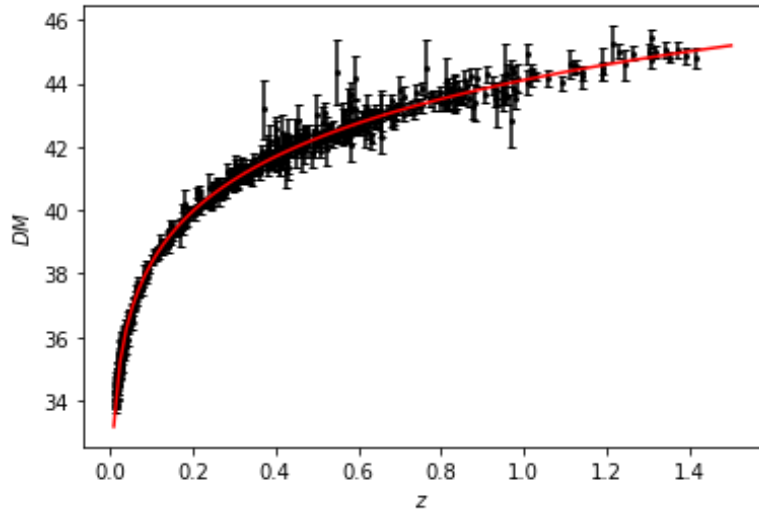
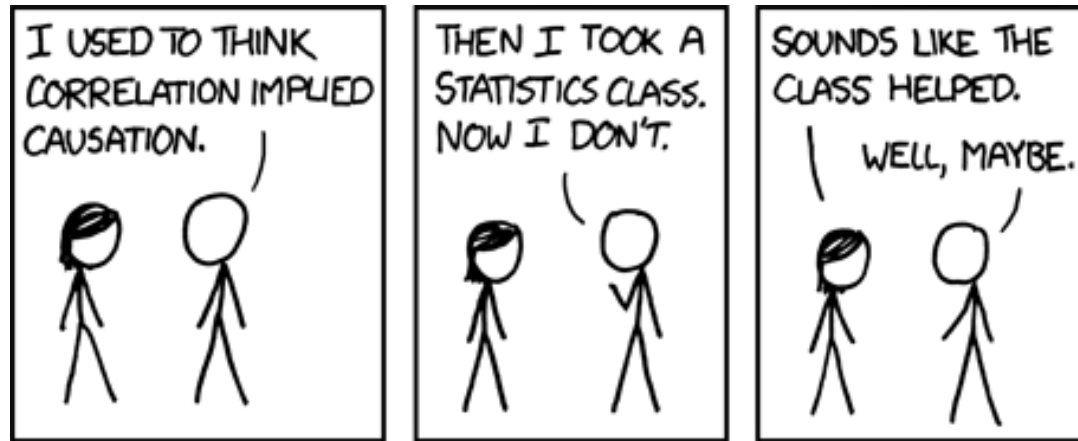


Class 3: Model-fitting

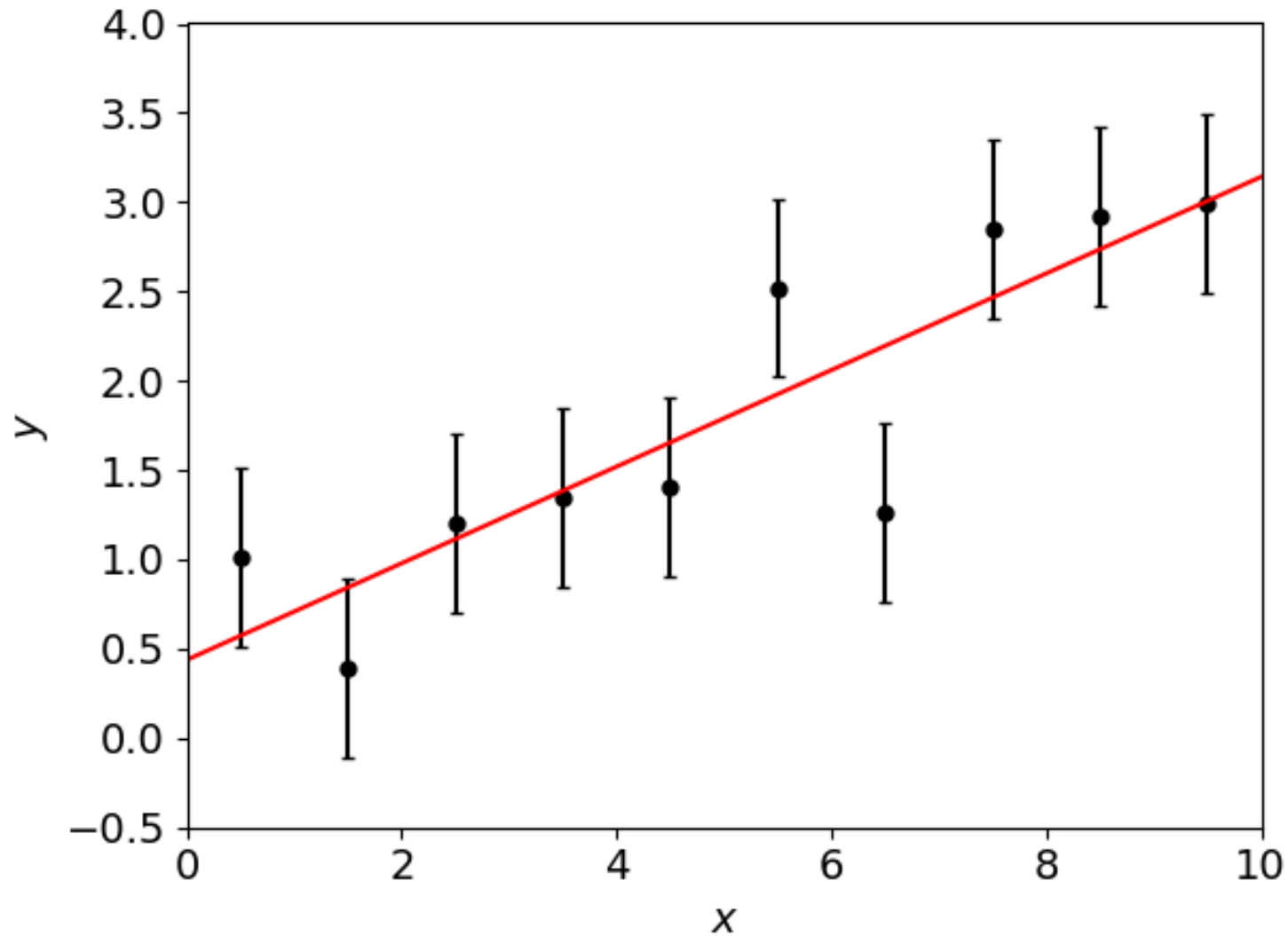


Class 3: Model-fitting



- Week 1: Probability and statistics
- Week 2: Correlation testing
- Week 3: Model fitting
- Week 4: Regression
- Week 5: Error estimates
- Week 6: Bayesian methods

Comparing data and models



Comparing data and models

Task	Situation	<i>Statistical question</i>

Comparing data and models

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Hypothesis testing	We have a set of N measurements $x_i \pm \sigma_i$, which a theorist says should have values μ_i	<i>How probable is it that these measurements would have been obtained, if the theory is correct?</i>

Comparing data and models

Task	Situation	<i>Statistical question</i>
Hypothesis testing	We have a set of N measurements $x_i \pm \sigma_i$, which a theorist says should have values μ_i	<i>How probable is it that these measurements would have been obtained, if the theory is correct?</i>
Parameter estimation	We have a parameterized model which describes the data, such as $y = ax + b$	<i>What are the best-fitting parameters and errors in those parameters?</i>

The χ^2 statistic

The most important statistic to help with these tasks is the χ^2 **statistic** between the data $x_i \pm \sigma_i$ and model μ_i

$$\chi^2 = \sum_{i=1}^N \left(\frac{x_i - \mu_i}{\sigma_i} \right)^2$$

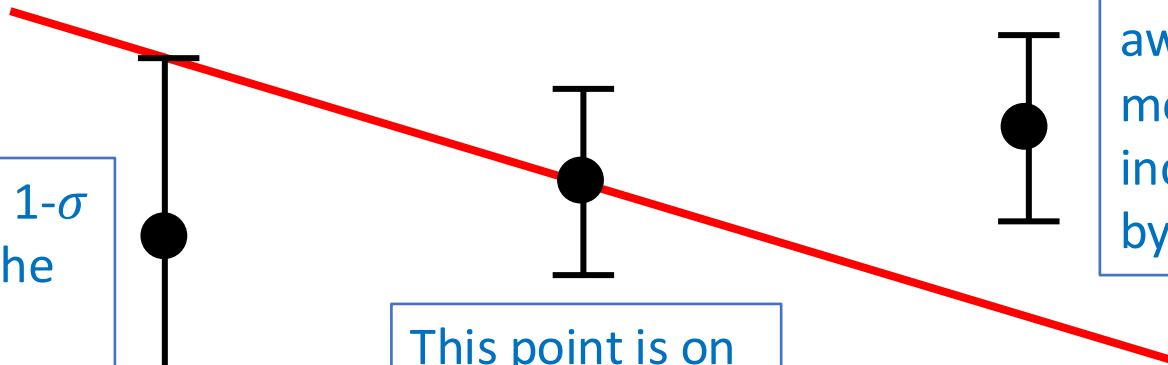
χ^2 is a sum over the data points

This point is 1- σ away from the model, so increments χ^2 by 1.0

This point is on the model, so increments χ^2 by 0.0

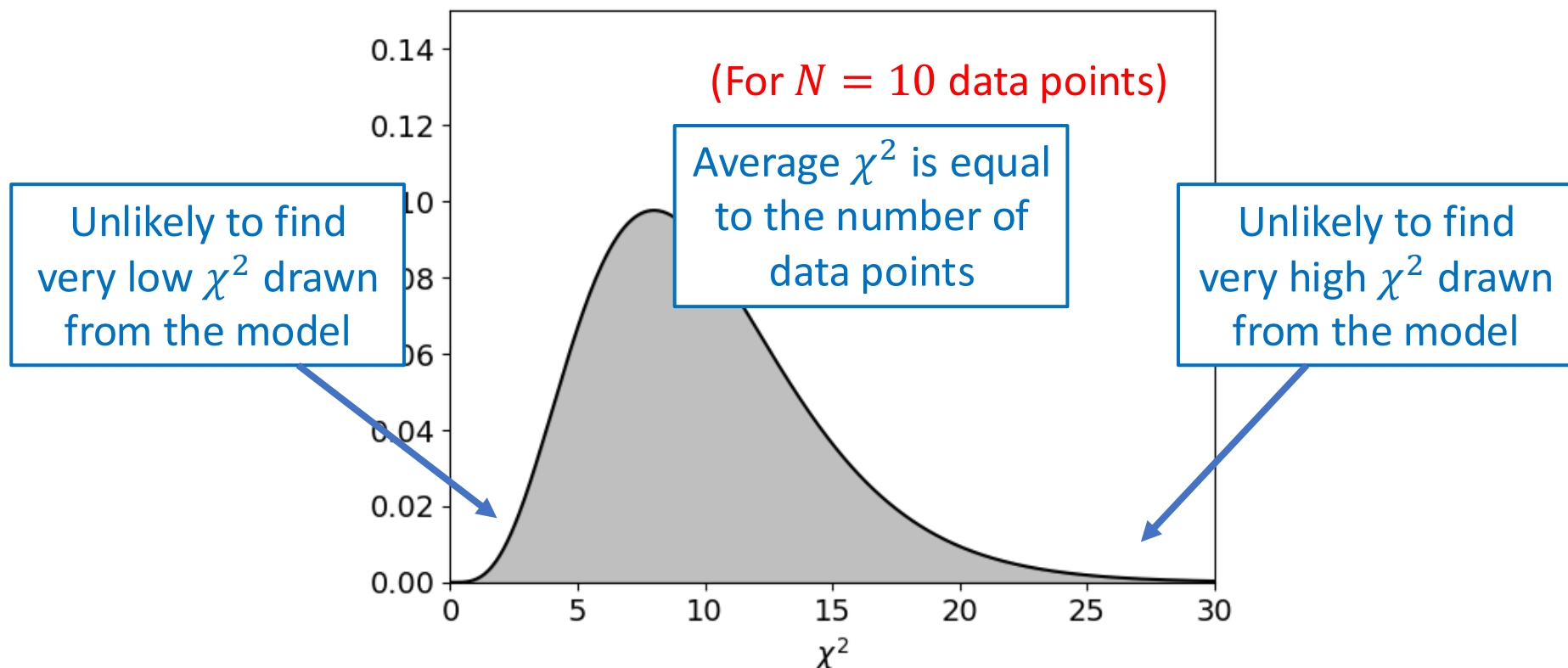
This point is 2- σ away from the model, so increments χ^2 by 4.0

Model



χ^2 probability distribution

Sampling many realizations of N data points from a particular model, using Gaussian statistics, the χ^2 statistic has a known **probability distribution**



χ^2 probability distribution

Sampling many realizations of N data points from a particular model, using Gaussian statistics, the χ^2 statistic has a known **probability distribution**

$$P(\chi^2) \propto (\chi^2)^{\frac{\nu}{2}-1} e^{-\chi^2/2}$$

- ν is the number of **degrees of freedom** (number of data points minus number of fitted parameters)
- The **mean** of the distribution: $\overline{\chi^2} = \nu$
- The **variance**: $\text{Var}(\chi^2) = 2\nu$
- If the model is correct, we expect $\chi^2 \sim \nu \pm \sqrt{2\nu}$

χ^2 tests – examples

When testing a model, I find $\chi^2 = 2.3$ for $\nu = 20$ degrees of freedom. What action should I take?

When testing a model, I find $\chi^2 = 42.3$ for $\nu = 20$ degrees of freedom. What action should I take?

When testing a model, I find reduced $\chi^2/\nu = 1.5$. Can I reject the model?

I'm testing a model which predicts the number of objects in a series of bins, so I assign a Poisson error $\sqrt{N}$ for a χ^2 test. Why might this go wrong?

Use of χ^2 statistic as a hypothesis test

How do we use the χ^2 statistic to construct a **hypothesis test** for the “goodness of fit” between data and model?

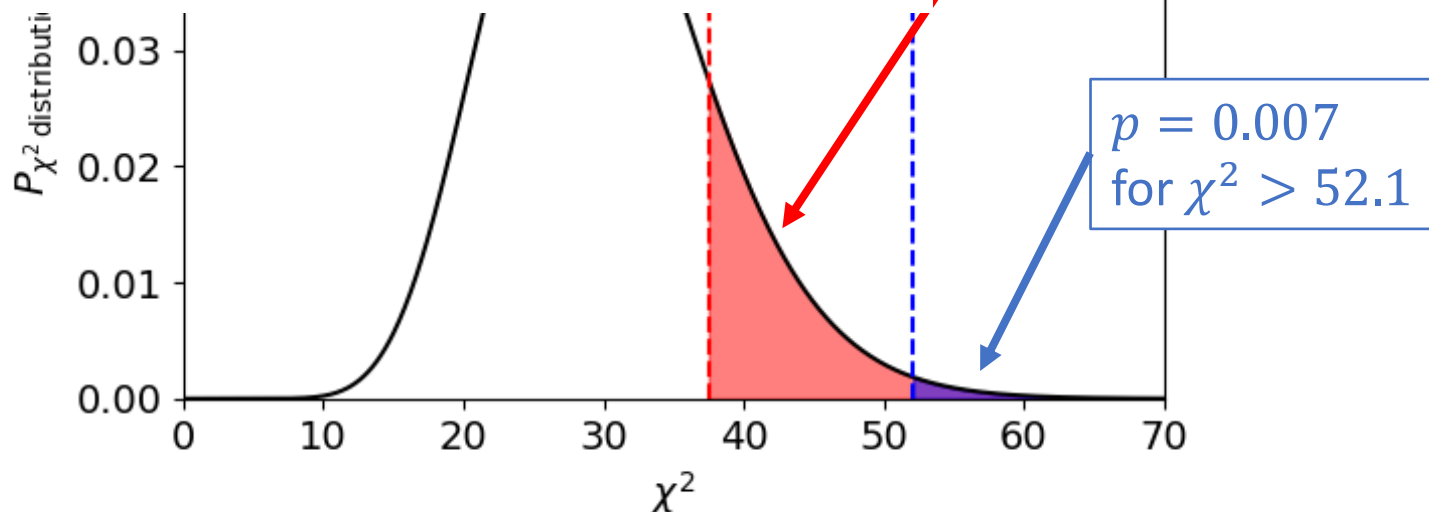
- **Null hypothesis:** the data are consistent with the model
- **Test statistic:** χ^2
- **Distribution of values:** The χ^2 probability distribution
- **Confidence statement:** What is the probability that this value of χ^2 , or a larger one, could arise by chance?
- If the ***p*-value is not low**, the data are consistent with the model, which is “*ruled in*”
- If the ***p*-value is low**, the model is “*ruled out*”

Worked example: χ^2 as a hypothesis test

- Suppose that $\nu = 30$ and we have 2 datasets with $\chi^2 = 37.5$ and $\chi^2 = 52.1$. What are the corresponding p -values?

```
: nu = 30 # number of degrees of freedom
chisq1 = 37.5 # 1st value of chi-squared to test
chisq2 = 52.1 # 2nd value of chi-squared to test
p1 = stats.chi2.sf(chisq1,nu) # probability of chi-squared exceeding
p2 = stats.chi2.sf(chisq2,nu) # probability of chi-squared exceeding
print('p-value for chi2 =',chisq1,'and nu =',nu,'is p =',p1)
print('p-value for chi2 =',chisq2,'and nu =',nu,'is p =',p2)
```

```
p-value for chi2 = 37.5 and nu = 30 is p = 0.16301345461145947
p-value for chi2 = 52.1 and nu = 30 is p = 0.007429708671565914
```



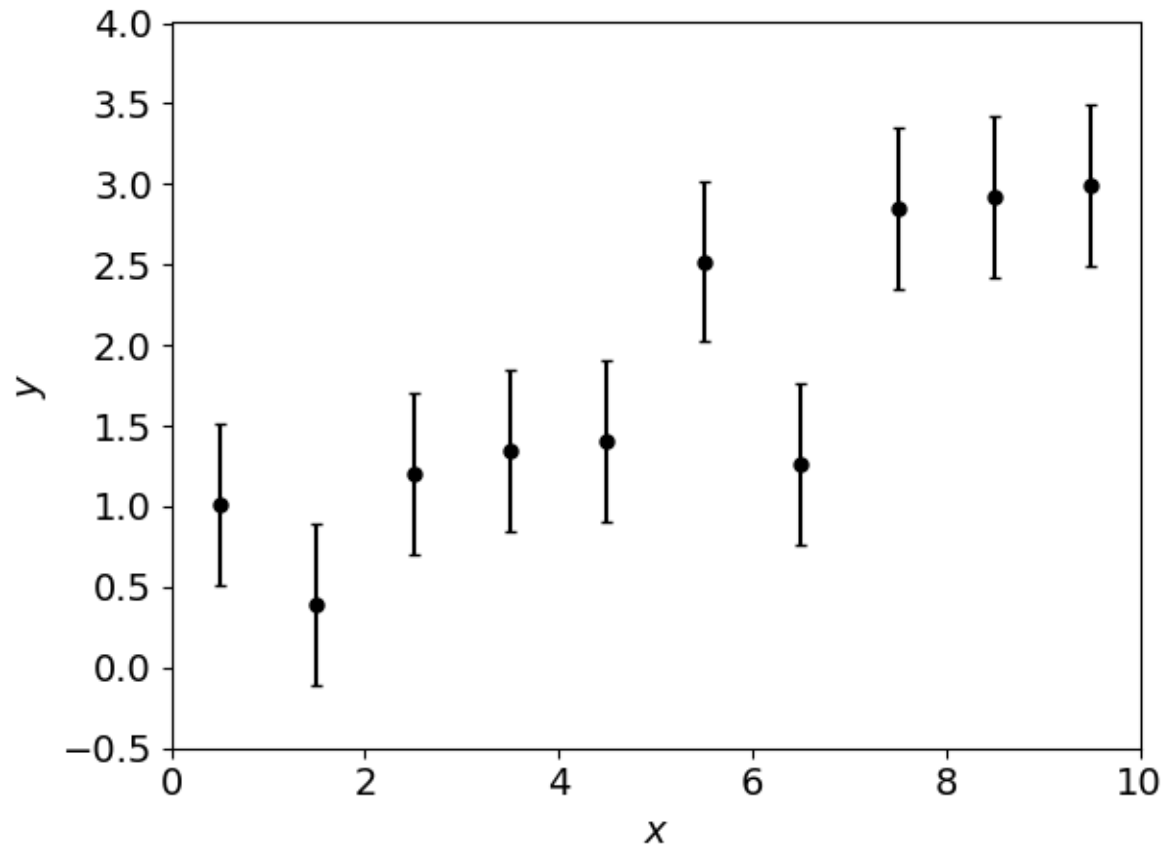
Use of χ^2 statistic for parameter fitting

How do we use the χ^2 statistic to determine the most likely values of **model parameters** and their error ranges?

- Suppose we are fitting a model with 2 free parameters (a, b)
- The most likely (“best-fitting”) values of (a, b) are found by **minimizing the χ^2 statistic**
- The joint error distribution of (a, b) can be found by **calculating the values of χ^2 over a grid of (a, b)** and enclosing a particular region $\chi^2 < \chi^2_{\min} + \Delta\chi^2$

Worked example: χ^2 for parameter fitting

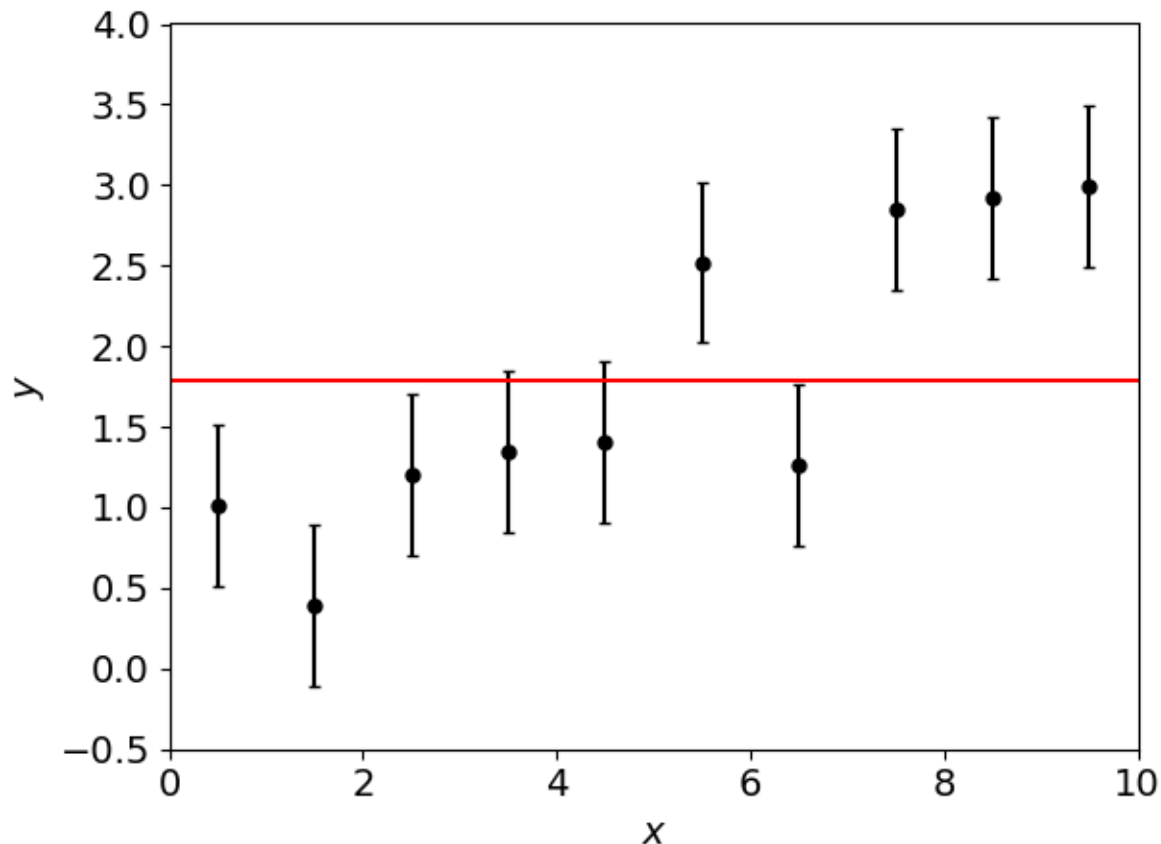
- Here is an example dataset containing $N = 10$ points:



Worked example: χ^2 for parameter fitting

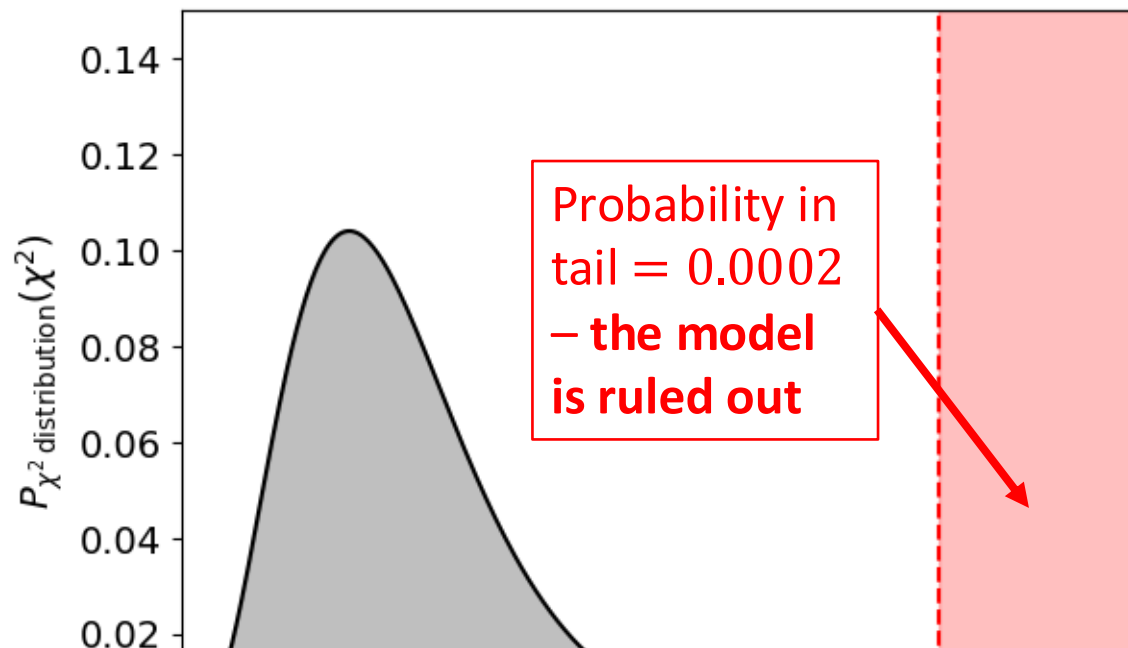
Could these points be fit by a constant $y = b$?

- Minimizing χ^2 , we find $\chi^2_{\min} = 31.6$ for $b = 1.79$



Worked example: χ^2 for parameter fitting

- Is the minimum χ^2 likely given the model?* Consider the χ^2 probability distribution for $\chi^2 > 31.6$ and $\nu = N - 1 = 9$:



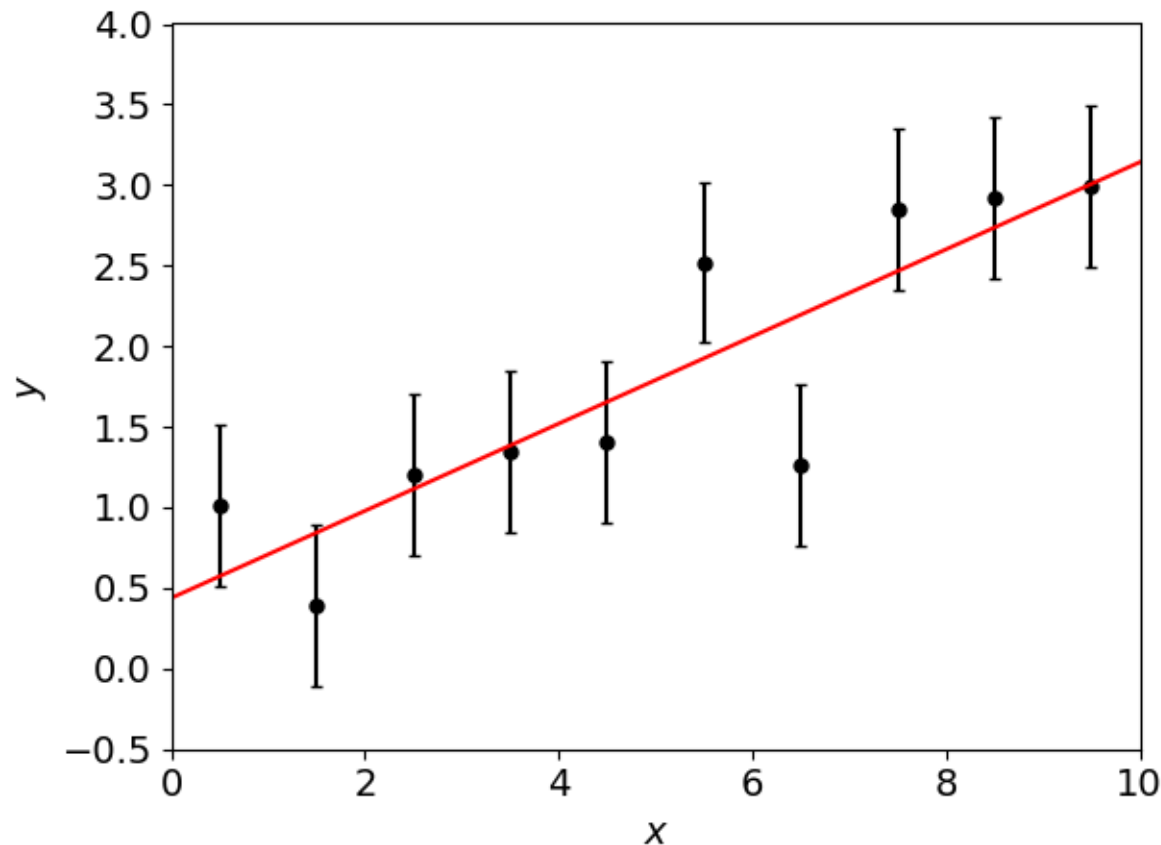
```
: # Probability of chi-squared to exceed the value we've found, if the data is sampled from the model
# Low values of this probability indicate that the data is unlikely to be drawn from the model
p1 = stats.chi2.sf(minchisq1,dof1)
print('p-value =',p1)
```

p-value = 0.00023020030084131723

Worked example: χ^2 for parameter fitting

Could these points be fit by a straight line $y = ax + b$?

- Minimizing χ^2 , we find $\chi^2_{\min} = 7.5$ for $a = 0.27$ and $b = 0.44$

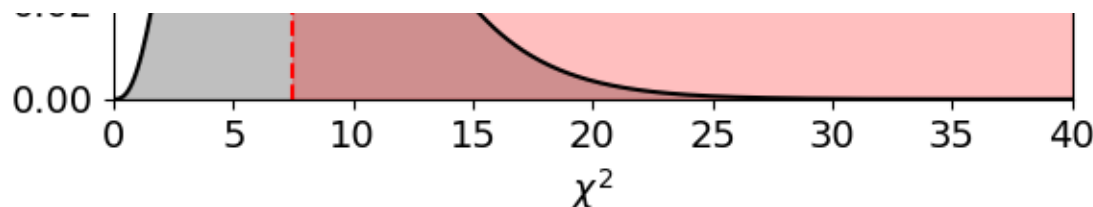


Worked example: χ^2 for parameter fitting

```
# Function which evaluates the chi-squared value of a model  $y=a*x+b$  for the (x,y) dataset
def chisqstraightline2(p,x,y,yerr):
    a,b = p[0],p[1]
    return np.sum(((y - (a*x + b))/yerr)**2)

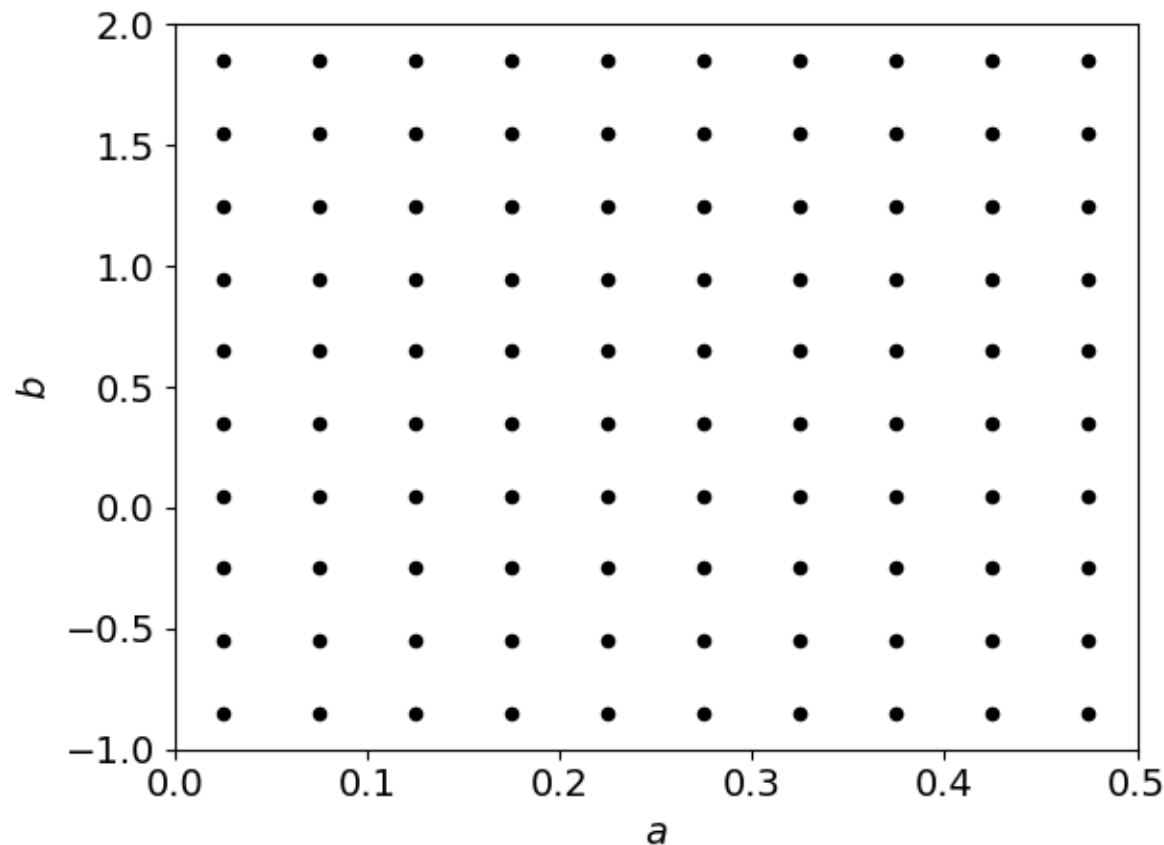
# Find the best fitting model  $y=a*x+b$  to the (x,y) dataset by minimising chi-squared
p0 = [0.2,1.] # initial guess of the values of a,b (result should not depend on this)
result = minimize(chisqstraightline2,p0,args=(x,y,yerr)) # perform the minimisation
a2,b2 = result['x'] # best-fitting values of a,b
minchisq2 = result['fun'] # chi-squared value corresponding to the best fit
dof2 = n-2 # number of degrees of freedom (data points minus fitted parameters)
p2 = stats.chi2.sf(minchisq2,dof2) # probability of chi-squared to exceed the value we've found
print('Best-fitting value of a =',a2)
print('Best-fitting value of b =',b2)
print('Minimum chi-squared =',minchisq2)
print('Number of degrees of freedom =',dof2)
print('p-value =',p2)
```

Best-fitting value of a = 0.27042421678017053
Best-fitting value of b = 0.4378789090155025
Minimum chi-squared = 7.502540606060829
Number of degrees of freedom = 8
p-value = 0.48350484463274246



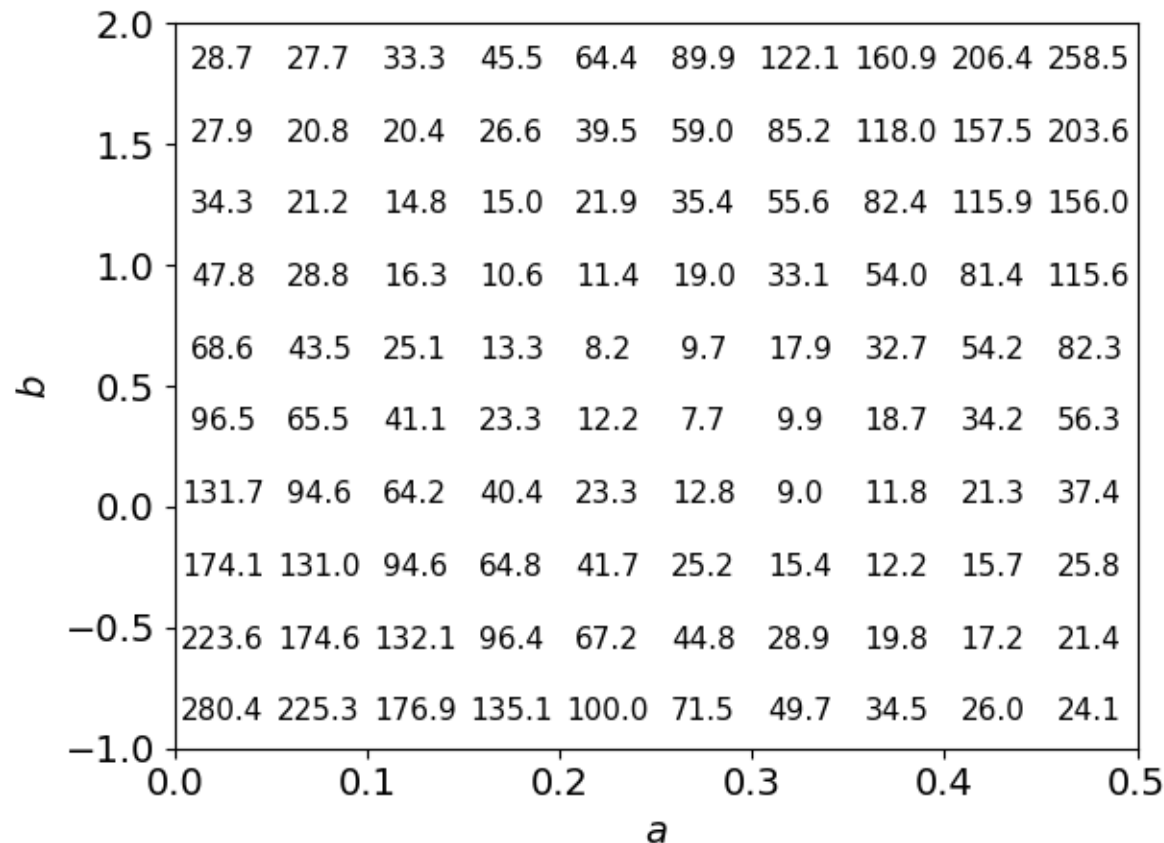
Worked example: χ^2 for parameter fitting

- *Now let's determine the error ranges.* What is the distribution of χ^2 values across the (a, b) grid?



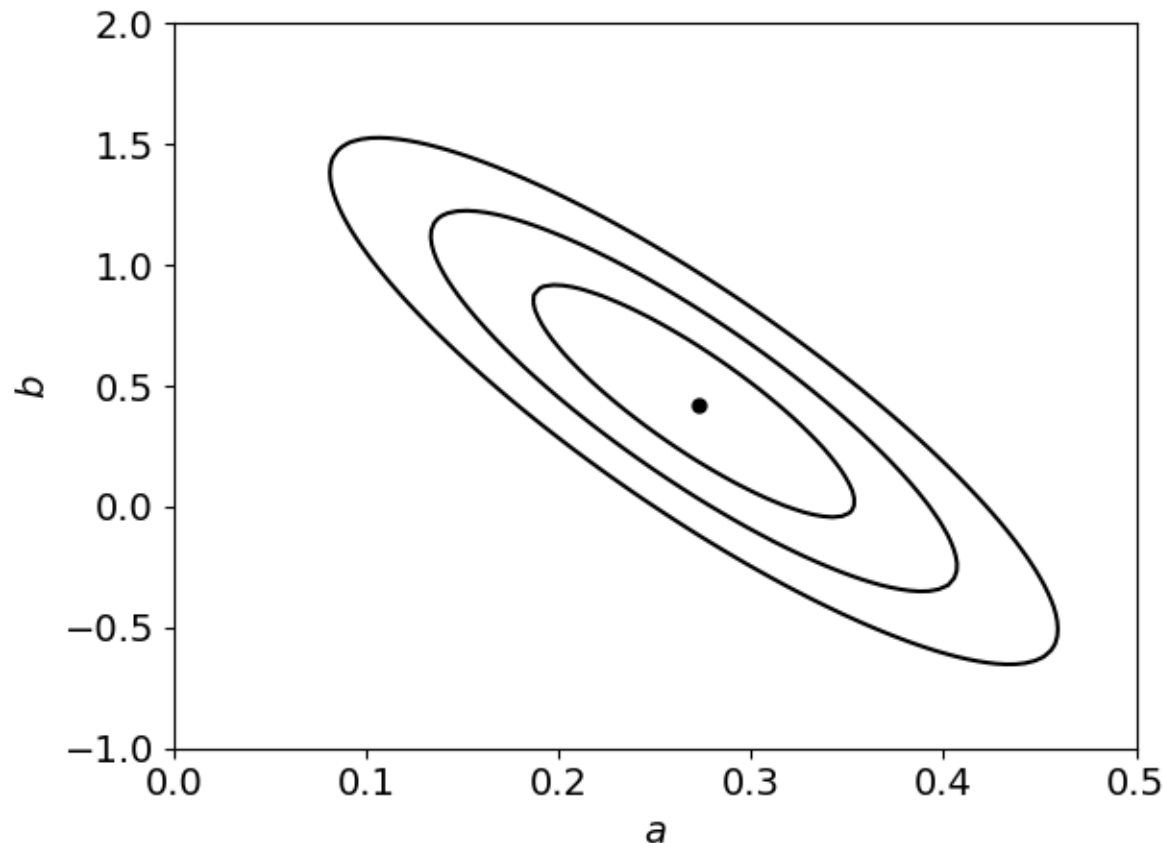
Worked example: χ^2 for parameter fitting

- *Now let's determine the error ranges.* What is the distribution of χ^2 values across the (a, b) grid?



Worked example: χ^2 for parameter fitting

- *Set confidence regions using $\Delta\chi^2$ intervals:* for 2 parameters, the (68,95,99)% regions are $\Delta\chi^2 = (2.30, 6.17, 11.8)$



Worked example: χ^2 for parameter fitting

- Set confidence regions using $\Delta\chi^2$ intervals: for 2 parameters, the (68,95,99)% regions are $\Delta\chi^2 = (2.30, 6.17, 11.8)$

$\Delta\chi^2$ as a Function of Confidence Level and Degrees of Freedom						
p	ν					
	1	2	3	4	5	6
68.3%	1.00	2.30	3.53	4.72	5.89	7.04
90%	2.71	4.61	6.25	7.78	9.24	10.6
95.4%	4.00	6.17	8.02	9.70	11.3	12.8
99%	6.63	9.21	11.3	13.3	15.1	16.8
99.73%	9.00	11.8	14.2	16.3	18.2	20.1
99.99%	15.1	18.4	21.1	23.5	25.7	27.8

2 parameters varying

1- σ
confidence

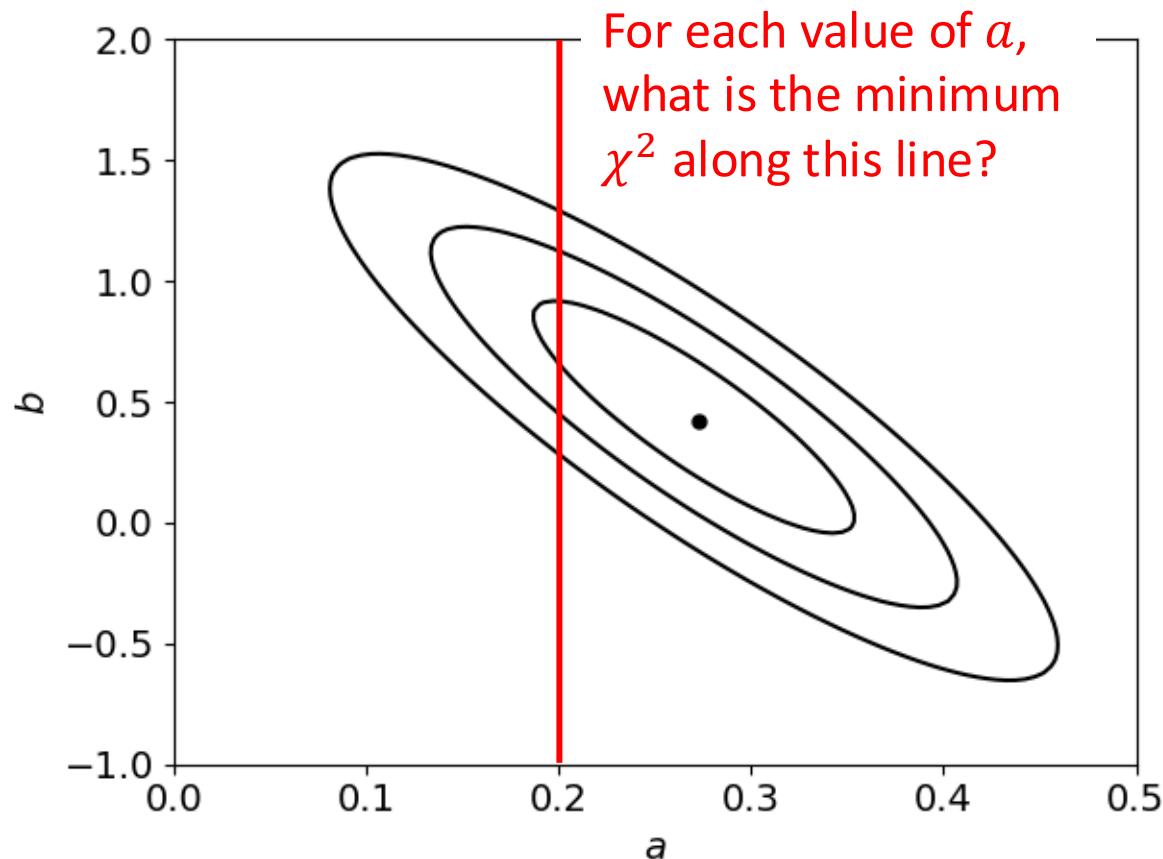
Worked example: χ^2 for parameter fitting

The joint error ranges when both parameters are varying are different from the error in each individual parameter

That's because the error in each individual parameter encompasses **all the possible values** of the other parameter(s) – this is called “**marginalisation over a parameter**”

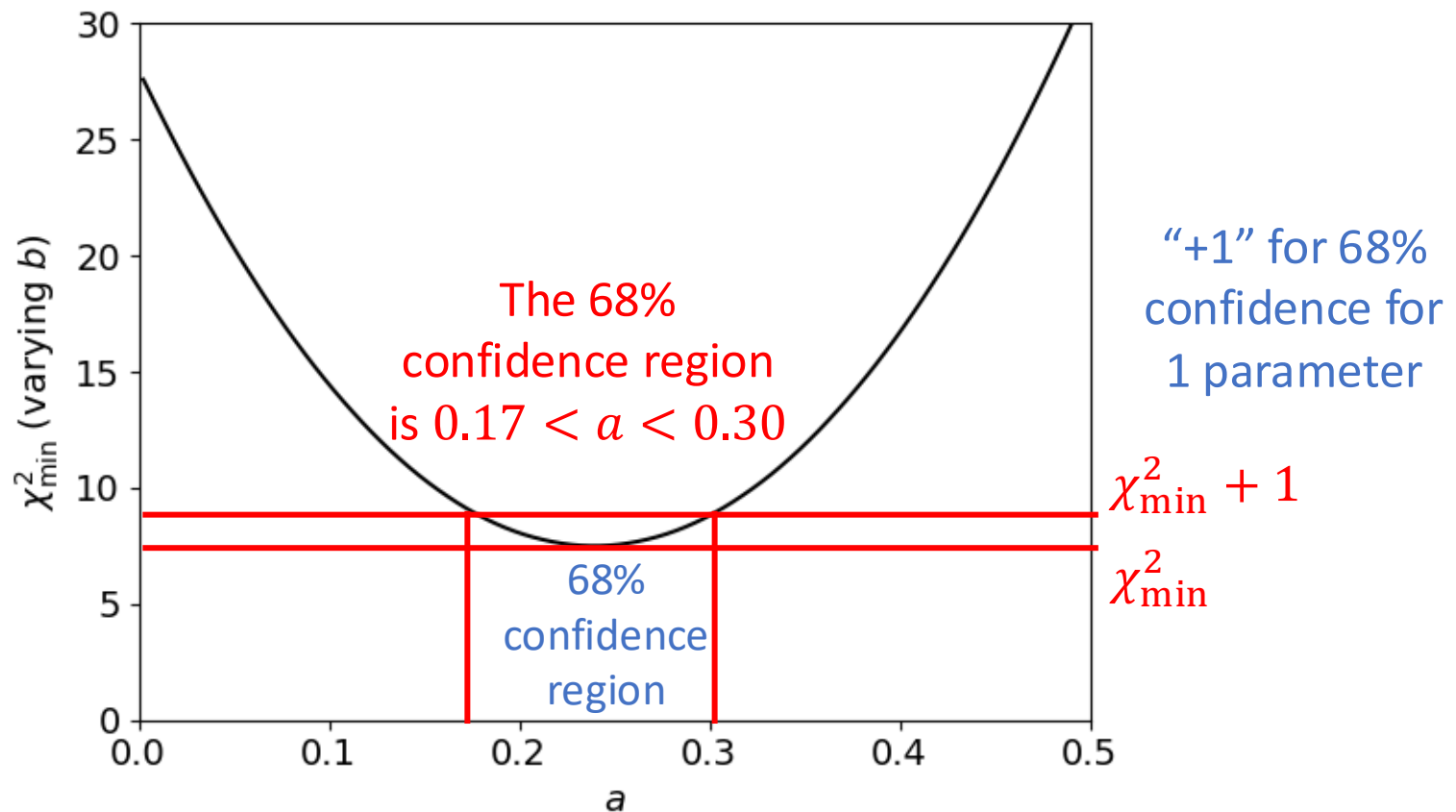
Worked example: χ^2 for parameter fitting

- *What about errors in individual parameters?* For each value of parameter a , find the minimum χ^2 varying parameter b :



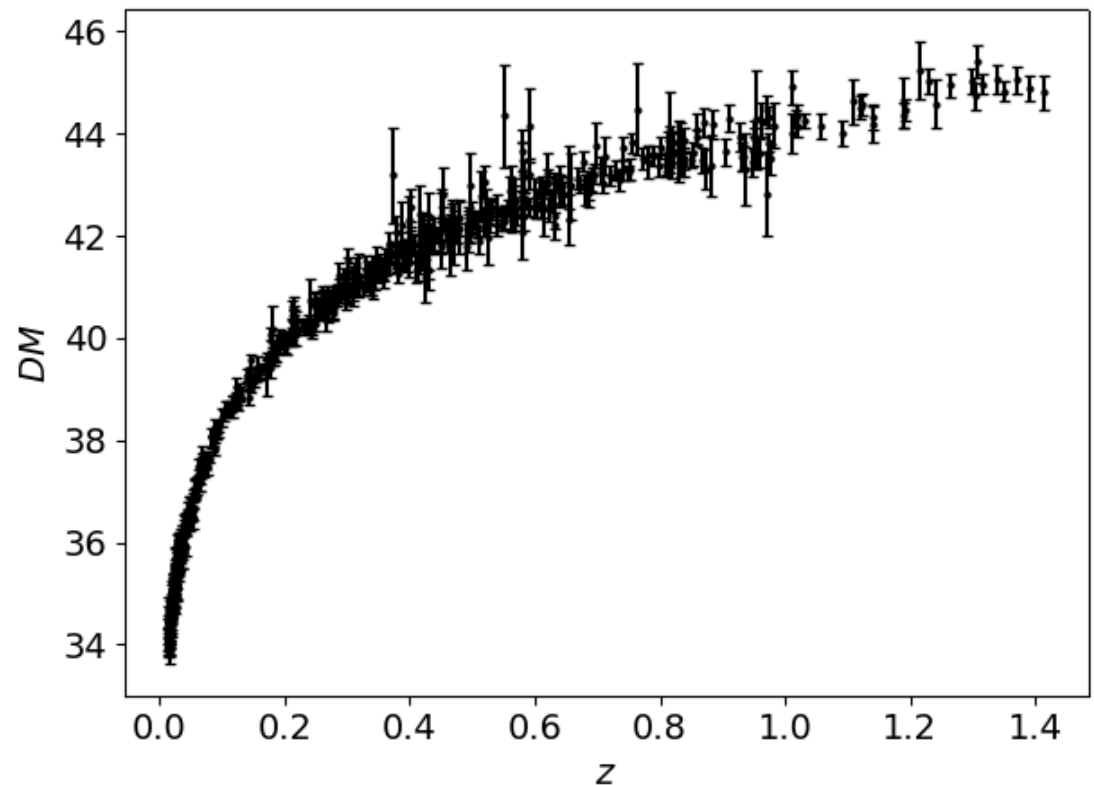
Worked example: χ^2 for parameter fitting

- What about errors in individual parameters?* For each value of parameter a , find the minimum χ^2 varying parameter b :



Activity: supernova cosmology

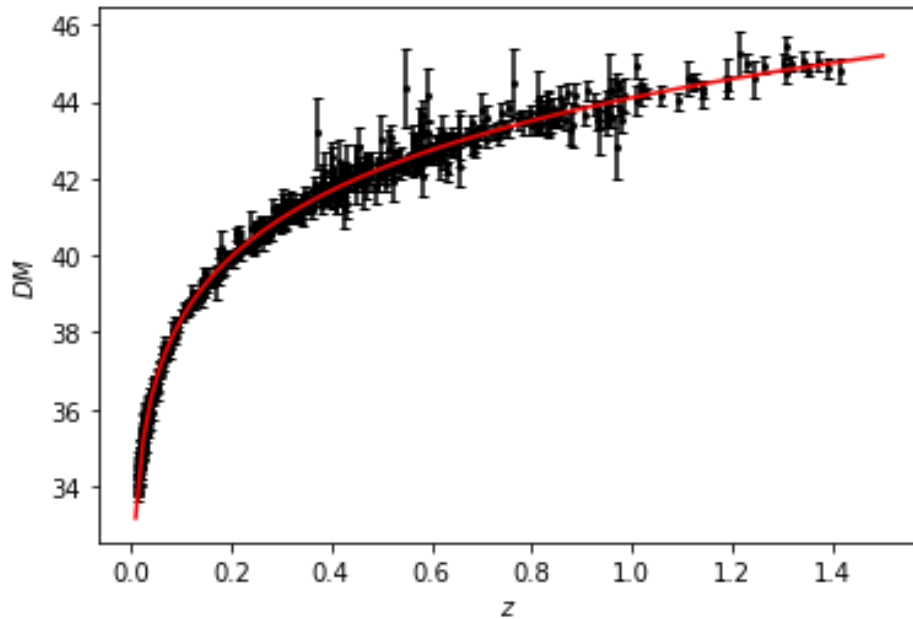
- In this Activity we will use a recent **supernova distance-redshift dataset** to determine the parameters $(\Omega_m, \Omega_\Lambda)$
- Minimize χ^2 and find the best-fitting $(\Omega_m, \Omega_\Lambda)$
- Construct the 2D confidence regions in $(\Omega_m, \Omega_\Lambda)$
- Determine the individual errors in Ω_m and Ω_Λ



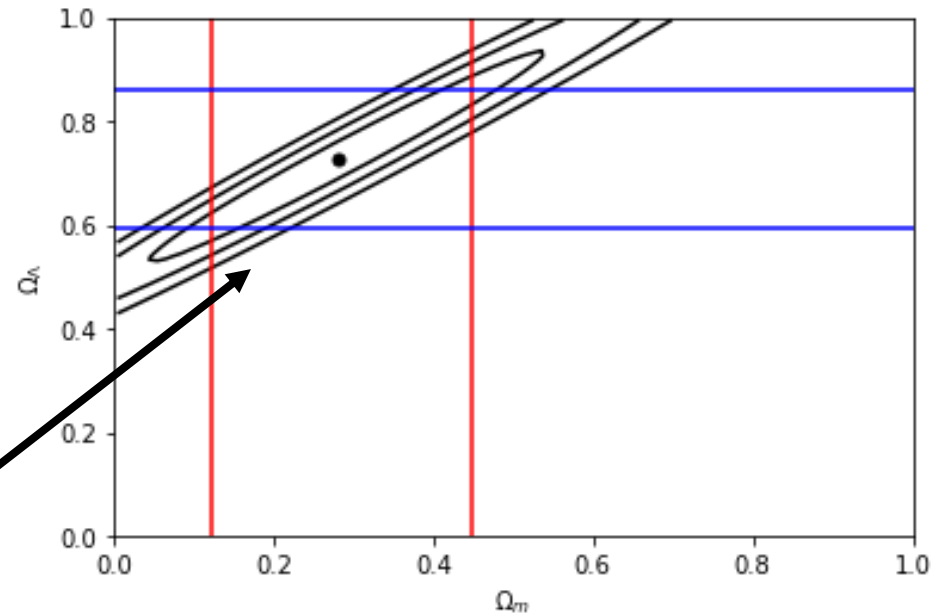
Activity: supernova cosmology

Best-fitting model:

$$\Omega_m = 0.28, \Omega_\Lambda = 0.73, \chi^2 = 562.2, \nu = 578$$



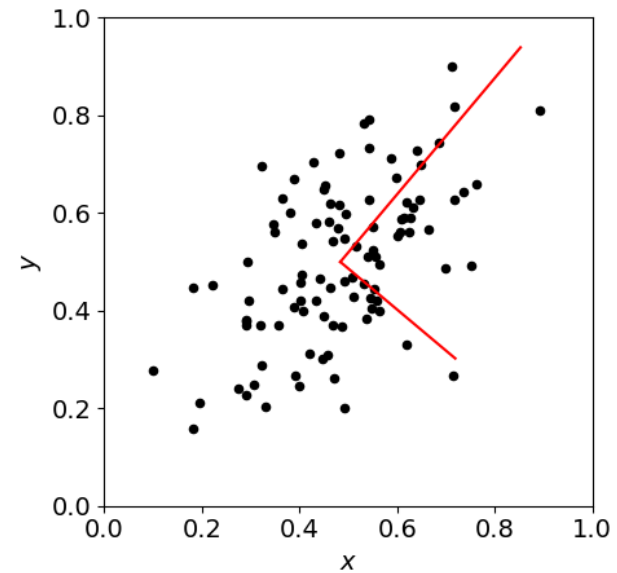
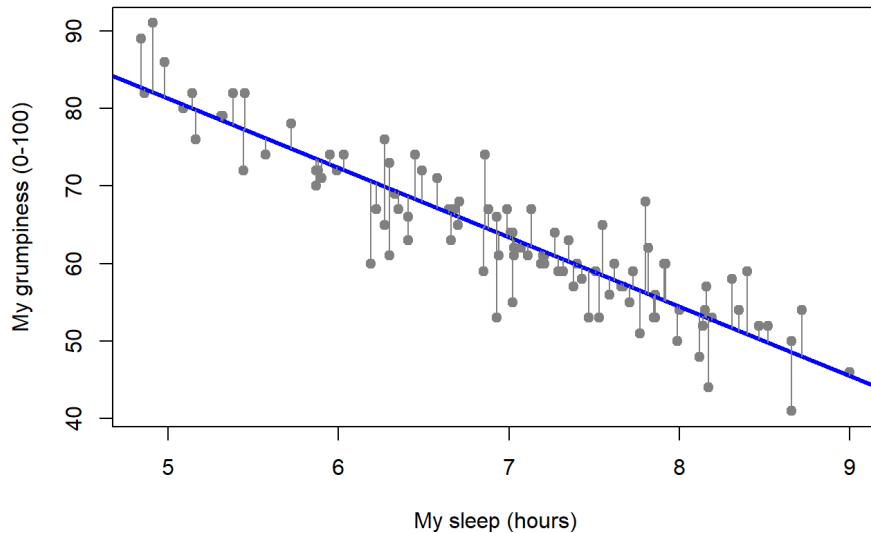
68% confidence for Ω_m : 0.12 – 0.45
68% confidence for Ω_Λ : 0.59 – 0.86



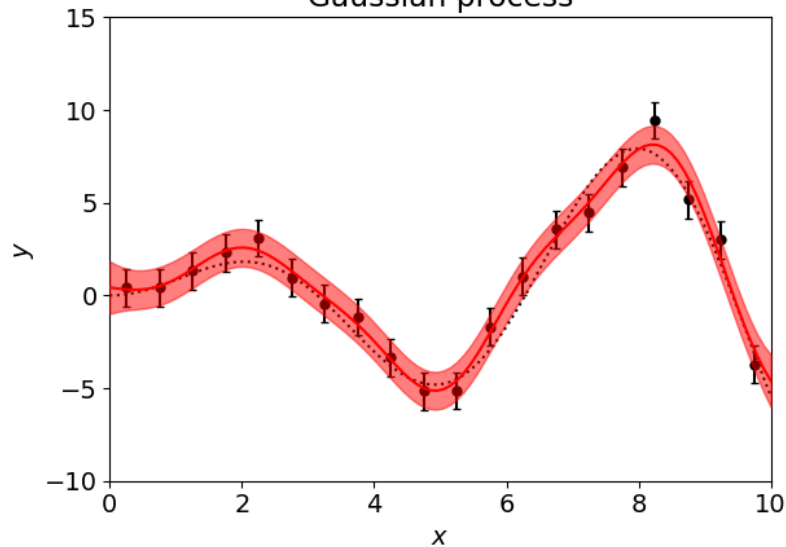
$\Delta\chi^2 = (2.30, 6.17, 11.8)$
for 2 parameters

Next week: Regression

Regression Line Close to the Data



Gaussian process



Cubic spline

